

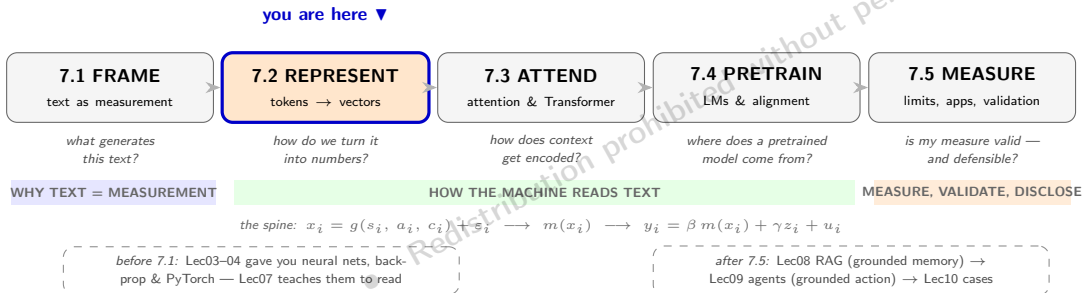
AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 7.2a: Preprocessing and Bag-of-Words

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Lecture 7 in One Map



Route: raw strings → clean text → tokens → counts (part 1/2 of REPRESENT — this deck builds the transparent count-based representations; part 2/2 turns counts into dense vectors).

From Strings to Tokens

Tokenization, subwords — and the Chinese special case

Tokenization: The First Step

- **Goal:** convert raw character sequences into discrete *tokens* that a neural network can consume.
- **Why not just split by spaces?**
 - Many economically relevant terms combine words, numbers, and symbols (e.g. S&P500, CO2, Q4_2025, stagflation)
 - Punctuation and formatting carry meaning ('\$1.2B' vs '1.2 billion')
 - Different languages have different word boundaries (or none at all)
- **What is a token?** A flexible unit:
 - A whole word ('inflation')
 - A subword ('infl', 'ation')
 - A single character or byte
 - A special marker (<EOS>, <PAD>, <USER>)

Vocabulary and the Tokenization Map

- Let A be the alphabet (Unicode characters). A tokenizer is a function:

$$\mathcal{K}: A^* \longrightarrow \{1, 2, \dots, M\}^*$$

mapping any text string into a sequence of **numbers** — each number is a positive integer that serves as an index into a lookup table of size M . The fact that the index is an integer is incidental; what matters is that it *uniquely identifies* each token, the same way a ZIP code uniquely identifies a location.

- Why integers?** Neural networks need numeric inputs; each token ID will index into an embedding lookup table.
- Practical constraint — context length:**
 - Models have a maximum input size n (typically 2,048 to 128,000 tokens; some 1M)
 - Long policy documents, FOMC transcripts, or 10-K filings often exceed n
 - Forces *chunking*, *summarization*, or *retrieval* (a hint at Lec08)
- Vocabulary sizes are typically 30,000–100,000 subword tokens. **To put this in perspective:** Merriam-Webster Unabridged lists $\approx 470,000$ entries; a college-educated native speaker actively uses roughly 20,000–40,000 words. A subword vocabulary of $M = 50,000$ therefore spans roughly $\frac{470,000}{50,000} \approx 9.4$ words.

Illustrative Example: “GDP rose by 2%”

- A simple tokenization might produce

(GDP, rose, by, 2, %)

mapping to integer IDs, e.g. (541, 823, 71, 312, 49).

- **Why integer mapping matters:**

- Each ID becomes the index of a learned embedding vector
- Numeric tokens ('2', '3', '5') get *their own* embeddings

- **Injectivity matters for forecasting:** a well-designed tokenizer maps

- “inflation rate is 3.5%”
- “inflation rate is 3.6%”

to *different* token sequences. Otherwise small but economically meaningful differences would collapse into the same input.

- **Tokenizer/model coupling:** the tokenizer is part of the model. If you swap the tokenizer, all the embedding rows are meaningless — this is why every published LLM ships with its tokenizer.

Subword Tokenization: BPE, WordPiece, SentencePiece

Algorithm	How it learns the vocabulary	Where you see it	Notable feature
Byte-Pair Encoding (BPE)	Start from chars; greedily merge the most frequent adjacent pair until vocab cap is reached	GPT-2/3/4, LLaMA, Mistral	Byte-level variant handles any Unicode
WordPiece	Like BPE, but merge the pair maximizing corpus likelihood instead of raw frequency	BERT, DistilBERT	## prefix marks subword continuations
SentencePiece	Trains directly on raw text (no whitespace presplit); unigram or BPE objective	T5, mT5, LLaMA, XLM-R	Multilingual; treats space as a normal character

Why subwords? OOV handling (QuantitativeEasing $\rightarrow$ Quant+itative+Easing); one token for common words, splits for rare/compound; one vocab across languages.

Cost: nowcasting may split into 3 tokens; 1.2345 into several digit-tokens — making numerical reasoning fragile.

Chinese Text: Segmentation Is the First Modeling Decision

English gets word boundaries for free; Chinese does not — so segmentation is itself a model.

- Chinese runs characters together; word boundaries must be *inferred*. Segmentation is a sequence-labeling problem: character by character, where does a word end?
- The classic ambiguity (romanized here): yanjiu-sheng-ming-ke-xue splits as
 - yanjiu | shengming kexue — “research on life science”, or
 - yanjiusheng | ming-kexue — “graduate student” + a nonsense remainder.

Economic text has its own: hangye zhengce zhichi reads as “industry-policy | support” or “industry | policy-support” — different head word, different meaning.

- **Tools:** Jieba, THULAC, LTP — each a dictionary plus a statistical/neural sequence labeler, trading speed against accuracy.
- **The economist’s trap — domain compounds:** terms like “monetary policy committee”, “macro-prudential framework”, “aggregate financing to the real economy” (shehui rongzi guimo) are not in general-purpose dictionaries; one mis-split changes the head word.
- **Fix:** load your domain glossary into the tool’s **user dictionary** so policy terms survive as whole tokens. Stopwords differ too: drop structural particles (de, le, shi); keep logical connectives (yinci “therefore”, ran’er “however”) for information-extraction tasks.

Preprocessing Is a Modeling Decision

Every cleaning switch changes the downstream estimate

Tokenization Is a Preprocessing Choice

- Tokenization sits inside a larger **preprocessing pipeline**:

$$x^{\text{raw}} \xrightarrow{\text{clean}} x^{\text{clean}} \xrightarrow{\text{tokenize}} \{t_{ij}\} \xrightarrow{\text{represent}} z_i,$$

from raw scraped text $\rightarrow$ normalized text $\rightarrow$ integer token IDs $\rightarrow$ downstream numeric representation (TF-IDF, embedding, ...).

- **Specification curve problem** (Denny & Spirling, 2018): just as economists report results under alternative lag orders, control sets, and sample restrictions, NLP results should be reported under alternative preprocessing pipelines.
 - LDA topics under different pipelines vary substantially — often by as much as the substantive variation of interest
 - Stemming and punctuation handling have the largest individual effects
 - **Lesson:** the pipeline is a modeling decision; report the full range as you would a VAR robustness table
- **Lesson for empirical economists:** report the *universe* of preprocessing choices. The “convenient” specification is not a result — it is a choice. This applies equally to tokenizer choice and vocabulary size.

The Seven Preprocessing Choices

- **Lowercasing** — map Fed and fed to the same token. Compresses vocabulary, but erases proper-noun signal.
- **Stemming / lemmatization** — cut suffixes so *banking/banks/banked* collapse to *bank*. Can erase tense, plurality, policy-language nuance.
- **Stop-word removal** — drop high-frequency function words. Reduces noise; but negation (*not*) and modality (*may, will*) can matter for central-bank text.
- **Punctuation handling** — keep, strip, or tokenize separately. Percent signs, decimals, sentence boundaries all matter.
- ***n*-gram inclusion** — treat “*yield curve*” as one token rather than two. Captures multi-word economic concepts at the cost of a larger vocabulary.
- **Frequency cutoffs** — drop terms in $< k$ or $> K$ documents. Removes typos and boilerplate; can also remove rare-but-informative events.
- **Language filtering** — restrict to English or apply per-language pipelines. Crucial for multilingual corpora like ECB speeches.

Each choice is an independent on/off switch $\Rightarrow 2^7 = 128$ pipelines $\Rightarrow$ report a *specification curve*, not a single run.

Thinkbox: One Boundary Flips the Sign

Where you draw token boundaries can reverse an economic conclusion.

FOMC-style sentence: "...growth is **not sufficient** to warrant further tightening."

- **Strategy A (unigrams):** not and sufficient are separate features. If sufficient loads positively in training, a bag-of-words model can extract a *positive* signal from a negation.
 - **Strategy B (bigrams):** not_sufficient is one feature — the negation survives as a unit.
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- Does the choice still matter under BERT-class models, whose attention can connect not to sufficient across positions? (Preview: mostly no — one reason Acts 3–4 exist.)
 - Either way: the tokenizer version and configuration belong in the replication package — they are part of the measurement instrument.

Thinkbox: Translate First, or Model the Original?

Comparing PBOC and Fed statements: two pipelines, two error structures.

Pipeline 1 — translate, then analyze.

Machine-translate the Chinese statements into English; score everything with one English model.

Cost: semantic drift. wenjian (“prudent/steady”) collapses to “prudent”; the policy-register difference between wenjian and shidu kuansong (“appropriately loose”) is flattened away.

Pipeline 2 — native text + multilingual model.

Extract representations in the original language (mBERT, XLM-R).

Cost: efficiency and depth. Chinese uses more tokens per unit of meaning, so context windows fill faster; multilingual models are weaker per-language than the best monolingual ones.

- If the conclusion hinges on terms that do not translate cleanly, run **both** pipelines and show the result survives.
- Declare the segmentation scheme and language strategy in the methods section — same status as the sample description.

Bag-of-Words and TF-IDF

The transparent baseline: count, weight, audit

Bag-of-Words: The Document–Term Matrix

The simplest representation: count words, forget order.

- Toy corpus — doc 1: “central bank cuts rates to support growth”; doc 2: “central bank raises rates to curb inflation”.

	central	bank	cuts	raises	rates	support	curb	growth	inflation
doc 1	1	1	1	0	1	1	0	1	0
doc 2	1	1	0	1	1	0	1	0	1

- In general: D documents $\times$ vocabulary of size V (typically $V \gg 10^4$); entry $X_{d,v}$ is binary, a raw count, or a TF-IDF weight (two slides ahead).
- **The assumption you buy:** “central bank cuts rates to support growth” = “growth supports central bank rate cuts”. Harmless when counting topic words (EPU); fatal for direction, causation, negation.
- **Two built-in statistical consequences:**
 - **Sparsity:** a document uses 1–2% of the vocabulary $\Rightarrow$ the matrix is >99% zeros; with $D \ll V$, OLS on raw counts is not even identified — dimension reduction (PCA, topic models) or regularization (Lasso) is mandatory.

Term Frequency and Inverse Document Frequency

Weight words by how much they distinguish a document — not by how often they appear.

TF — share of document d that is word w :

$$\text{TF}(w, d) = \frac{\text{count}(w, d)}{\text{total words in } d}$$

“the Fed raised rates because inflation persisted” (7 words): every word gets $\text{TF} = 1/7 \approx 0.14$ — the scores as high as inflation.

Limitation: TF alone rewards frequency, not information. We need a counterweight.

IDF — rarity of w across N documents:

$$\text{IDF}(w) = \log \left(\frac{N}{\#\{\text{docs containing } w\}} \right)$$

Corpus of $N = 100$ FOMC speeches:

word	docs	IDF
<i>the</i>	100	0
<i>inflation</i>	60	0.51
<i>yield curve</i>	5	3.00

Rare $\Rightarrow$ informative $\Rightarrow$ up-weighted. *the* zeroes out automatically — no stop-word list needed.

TF-IDF: The Workhorse Count Representation

Multiply the two and stack across the vocabulary: each document becomes a sparse vector in $\mathbb{R}^{|V|}$.

$$\text{TF-IDF}(w, d) = \text{TF}(w, d) \cdot \text{IDF}(w)$$

Toy 3-speech matrix (illustrative):

	<i>the</i>	<i>inflation</i>	<i>yield curve</i>
speech A (hawkish)	0	0.08	0.00
speech B (dovish)	0	0.05	0.00
speech C (technical)	0	0.02	0.21

Properties:

- Down-weights ubiquitous words automatically — boilerplate like “financial statements” gets weight ≈ 0
- Highlights each document's distinctive vocabulary
- Still sparse, still orthogonal: *central bank* and *Federal Reserve* remain unrelated axes $\Rightarrow$ the motivation for dense embeddings (T2b)

Time-series look-ahead bias. If IDF is computed on the *full sample*, period- t weights peek at future documents — unacceptable in forecasting or causal designs. Compute IDF on a rolling window ending at $t - 1$; watch the ECB Word2Prices design (T5b) do exactly this. A side effect either way: if direction words (*not*, *up*, *down*) appear in nearly every document, IDF crushes their weight — yet they carry the policy signal. Protect them explicitly.

When the Bag of Words Is Still the Right Tool

Orthogonality is a real defect — and yet counts win whole classes of research problems.

- **When the target is “how often do well-defined words appear”:** EPU (Thread A), the Loughran–McDonald finance dictionary, Hassan et al.’s bigram political risk — all count-based *by design*.
- **The killer feature is auditability:** you can list exactly which article and which phrase triggered the index — line-by-line external validation. No embedding offers that evidence chain.
- **Also the sensible default when:** the corpus is small ($D < 1,000$); compute is tight; or the measure must plug into a standard econometric pipeline with transparent features.

Choose the representation for the *role* the text plays (T1’s three roles): measurement wants transparency first; behavior and nuance want context — which is exactly where T2b and Act 3 are headed.

Bridge: Counts Cannot See Synonyms

The bag of words has taken us as far as orthogonal axes can go.

- In the document-term matrix, *hawkish* and *tightening* are perpendicular; *unemployment* and *joblessness* share nothing. Every count-based measure inherits this blindness — the recall problem behind all dictionary methods.
- **Next deck:** learn coordinates *from co-occurrence*, so related words end up near each other — Word2Vec, GloVe, and the geometry that lets us do arithmetic with meaning.
- The DGP lens carries over unchanged: an embedding is just a richer $m(\cdot)$; the hazards a_i , c_i do not vanish because the vectors get denser.

Arc: T1 FRAME $\rightarrow$ T2 REPRESENT $\rightarrow$ T3 ATTEND $\rightarrow$ T4 PRETRAIN $\rightarrow$ T5 MEASURE.